

Jonas Conneryd

Department of Computer Science
Lund University
Lund, Sweden

Date of birth: July 31st, 1997
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Education

Lund University

Ph.D. in Computer Science

Advisor: Prof. Tatyana Turova

Thesis: *On the Algebraic Proof Complexity of Constraint Satisfaction*

Opponent: Prof. Alexander Razborov, University of Chicago

Lic. Eng. in Computer Science

Thesis: *On the Average-Case Proof Complexity of Graph Coloring*

Opponent: Prof. Nutan Limaye, ITU Copenhagen

KTH Royal Institute of Technology

M.Sc. in Mathematics (joint with Stockholm University)

Thesis: *Geometric Bounds for Steklov Eigenvalues on Graphs* (awarded Mittag-Leffler Prize)

B.Sc. in Engineering Physics

Thesis: *Explicit Symplectic Integrators for Non-Separable Hamiltonians in Molecular Dynamics*

Lund, Sweden

Thesis defended September 2026

Thesis defended May 2025

Stockholm, Sweden

June 2021

June 2019

Research Interests

Computational complexity theory, particularly proof complexity with an algebraic flavor and the complexity of (random) constraint satisfaction problems.

Publications

- [1] Jonas Conneryd. *On the Algebraic Proof Complexity of Graph Coloring*. Ph.D. thesis. Lund University, 2026.
- [2] Jonas Conneryd, Yassine Ghannane, Jakob Nordström, Shuo Pang, Kilian Risse, and Dmitry Sokolov. *On the Power of Polynomial Calculus over Non-Boolean Domains*. 2026.
- [3] Jonas Conneryd, Yassine Ghannane, and Shuo Pang. *Lower Bounds for CSP Hierarchies Through Ideal Reduction*. SODA 2026.
- [4] Jonas Conneryd, Susanna F. de Rezende, Jakob Nordström, Shuo Pang, and Kilian Risse. *Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz*. FOCS 2023.
- [5] Jonas Conneryd. *Geometric Bounds for Steklov Eigenvalues on Graphs*. M.Sc. thesis. Stockholm University, 2021.

Talks

McGill University

Lower Bounds for CSP Hierarchies Through Ideal Reduction

Montréal, QC, Canada

Summer 2026

ACM-SIAM Symposium on Discrete Algorithms (SODA)

Lower Bounds for CSP Hierarchies Through Ideal Reduction

Vancouver, BC, Canada

Winter 2026

KTH Royal Institute of Technology

Lower Bounds for CSP Hierarchies Through Ideal Reduction

Stockholm, Sweden

Fall 2025

Oxford Proof Complexity Workshop

Lower Bounds for CSP Hierarchies Through Ideal Reduction

Oxford, United Kingdom

Summer 2025

Proof Complexity and Beyond, Oberwolfach Workshop 2413

Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz

Oberwolfach, Germany

Spring 2024

Algorithmic Research Cooperation around Øresund (ARCO)

Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz

Odense, Denmark

Fall 2023

IEEE Symposium on Foundations of Computer Science (FOCS)

Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz

Santa Cruz, CA, USA

Fall 2023

Research Visits and Workshops

McGill University

Month-long visit to the group of [Asst. Prof. Robert Robere](#).

Montréal, QC, Canada

Summer 2026

Swiss Winter School on Theoretical Computer Science

Visiting researcher for a [winter school](#) jointly organized by [EPFL](#) and [ETH Zürich](#).

Zinal, Switzerland

Winter 2026

KTH Royal Institute of Technology

Month-long visit to the theory group.

Stockholm, Sweden

Fall 2025

University of Oxford

Visiting researcher for the workshop *Proof Complexity* in 2024 and 2025.

Oxford, United Kingdom

Summers 2024, 2025

Centre International de Rencontres Mathématiques

Visiting researcher for the *Complexity as a Kaleidoscope* research school.

Marseille, France

Spring 2025

DIMACS, Rutgers University

Visiting researcher for the *Frontiers in Complexity Workshop*.

New Brunswick, NJ, USA

Summer 2024

McGill University

Week-long visit to the group of Asst. Prof. Robert Robere.

Montréal, QC, Canada

Summer 2024

Mathematisches Forschungsinstitut Oberwolfach

Visiting researcher for the workshop *Proof Complexity and Beyond*.

Oberwolfach, Germany

Spring 2024

Chalmers University of Technology

Visiting researcher for the *Workshop on Algebra and Computation*.

Gothenburg, Sweden

Summer 2023

Simons Institute for the Theory of Computing at UC Berkeley

2-month visiting graduate student for the program *Satisfiability: Extended Reunion*.

Berkeley, CA, USA

Spring 2023

Schloss Dagstuhl

Visiting researcher for the workshop *Theory and Practice of SAT and Combinatorial Solving*.

Dagstuhl, Germany

Fall 2022

Honors and Scholarships

- Wallenberg Foundation and WASP International Postdoctoral Scholarship (ca 100 000 EUR); 2026
- WASP travel grant for one-month visit to McGill University (40 000 SEK); 2026
- Oberwolfach Leibniz Graduate Student; Mathematisches Forschungsinstitut Oberwolfach, 2024
- WASP travel grant for two-month visit to Simons Institute at UC Berkeley (80 000 SEK); 2023
- Accepted as *WASP* affiliated student; 2021
- Mittag-Leffler Prize for outstanding M.Sc. theses in mathematics; Stockholm University, 2021
- Ingenjör Ernst Johnson Scholarship for outstanding academic achievements; KTH, 2020, 2021
- Henrik Göransson Sandviken Scholarship for outstanding academic achievements; KTH, 2018
- University Merit Scholarship for outstanding academic achievements; KTH, 2018, 2019, 2020, 2021

Service

Reviewed papers for the following venues:

- Conferences: FOCS '22; CCC '22; CCC '23; SAT '23; SAT '24; ICALP '24 (x2); AAI '23; AAI '24; SOSA '25; STOC '25; FOCS '26
- Journals: ACM Transactions on Computational Logic; Theoretical Computer Science; SIAM Journal on Discrete Mathematics

Teaching Experience

Lund University

- EDAA40/75 Discrete Structures in Computer Science, 2022, 2024, 2025, 2026
- EDAN55 Advanced Algorithms, 2023, 2024
- EDAN01 Constraint Programming, 2022, 2023

KTH Royal Institute of Technology

- SF1661 Perspectives on Mathematics, 2018
- SF1624 Algebra and Geometry, 2017

Work Experience

AP3 Third National Swedish Pension Fund

ILS Intern

Quantitative analysis of insurance-linked securities (ILS).

Stockholm, Sweden

2019–2021

Technical Skills

Programming: Python, Go, $\text{\LaTeX}$, Julia, MATLAB

Software: RMS Miu